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Machu Picchu

Machu Picchu[a] is a 15th-century Inca citadel located in the Eastern Cordillera of southern Peru on a mountain ridge at 2,430 meters (7,970 ft).[8] Often referred to as the "Lost City of the Incas",[9] it is the most familiar icon of the Inca Empire. It is located in the Machupicchu District within Urubamba Province[10] above the Sacred Valley, which is 80 kilometers (50 mi) northwest of Cusco. The Urubamba River flows past it, cutting through the Cordillera and creating a canyon with a subtropical mountain climate.[11]

The Inca civilization had no written language and following the first encounter by the Spanish soldier Baltasar Ocampo, no Europeans are recorded to have visited the site from the late 16th century until the 19th century. As far as historical knowledge extends, there are no existing written records detailing the site during its period of active use. The names of the buildings, their supposed uses, and their inhabitants, are the product of modern archaeologists based on physical evidence, including tombs at the site. Machu Picchu was built in the classical Inca style, with polished dry-stone walls. Its three primary structures are the *Temple of the Sun*, the *Temple of the Three Windows,* and the *Intihuatana*. Most of the outlying buildings have been reconstructed in order to give visitors a better idea of how they originally appeared.[12] By 1976, 30 percent of Machu Picchu had been restored[12] and restoration continues.[13] Most recent archaeologists believe that Machu Picchu was constructed as an estate for the Inca emperor Pachacuti (1438–1472). The Inca built the estate around 1450 but abandoned it a century later, at the time of the Spanish conquest. According to the new AMS radiocarbon dating, it was occupied from c. 1420–1532.[14] Historical research published in 2022 claims that the site was probably called Huayna Picchu by the Inca people themselves, as it exists on the smaller peak of the same name.[15][16]

Machu Picchu was declared a Peruvian Historic Sanctuary in 1982 and a UNESCO World Heritage Site in 1983.[8] In 2007, Machu Picchu was voted one of the New Seven Wonders of the World in a worldwide internet poll.[17]

Machu Picchu

Machu Picchu

Shown within Peru

Click on the map for a fullscreen view

Location	Cusco Region, Peru
Coordinates	13°09′48″S 72°32′44″W
Altitude	2,430 m (7,972 ft)
History	
Founded	c. 1450
Abandoned	c. 1532 – c. 1565[1][2]
Cultures	Inca civilization
Site notes	
Discovered	Agustín Lizárraga (1902) Hiram Bingham (1911)
Website	https://www.machupicchu.gob.pe/?lang=en
UNESCO World Heritage Site	
Official name	Historic Sanctuary of Machu Picchu
Location	Latin America and the Caribbean
Criteria	Mixed: i, iii, vii, ix
Reference	274 (https://whc.unesco.org/en/list/274)
Inscription	1983 (7th Session)
Area	38,160.87 ha

Etymology

The site is on a narrow saddle between two mountain peaks, Machu Picchu and Huayna Picchu. In the Quechua language, *machu* means "old" or "old person" and *wayna* means "young", while *pikchu* refers to a "summit," "peak," or "pyramid."[18][b] Thus, the name of the site is often translated as "old mountain" or "old peak".[19]

Although the original name given to the settlement by its builders is not definitively known, a 2021 study in *Ñawpa Pacha: Journal of the Institute of Andean Studies* suggests that the site was likely called "Huayna Picchu," after the smaller peak nearby, or simply "Picchu."[15][20] According to the research, the association of the name Machu Picchu with the ruins likely began with American explorer Hiram Bingham's 1911 publications, a conclusion supported by Bingham's field notes, early maps, and historical documents.[15]

History

Photo by Hiram Bingham III in 1912 after major clearing and before reconstruction work began[12][13]

Machu Picchu was previously believed (by Richard L. Burger, professor of anthropology at Yale University) to have been built in the 1450s.[21] However, a 2021 study led by Burger used radiocarbon dating (specifically, AMS) to reveal that Machu Picchu may have been occupied from around 1420 to 1530 AD.[22][23] Construction appears to date from two great Inca rulers, Pachacutec Inca Yupanqui (1438–1471) and Túpac Inca Yupanqui (1472–1493).[24][25]:xxxvi

A consensus among archaeologists is that Pachacutec ordered the construction of the royal estate for his use as a retreat, most likely after a successful military campaign. Although Machu Picchu is considered to be a "royal" estate, it would not have been passed down in the line of succession. Rather it was used for 80 years before being

abandoned, seemingly because of the Spanish conquests in other parts of the Inca Empire.[21] It is possible that most of its inhabitants died from smallpox introduced by travelers before the Spanish conquistadors even arrived in the area.[26]

Ancient life

Daily life in Machu Picchu

View of the ancient houses

During its use as an estate, it is estimated that about 750 people lived there, with most serving as support staff (*yanaconas, yana*)[27][28] who lived there permanently. Though the estate belonged to Pachacutec, religious specialists and temporary specialized workers (*mayocs*) lived there as well, most likely for the ruler's well-being and enjoyment. During the harsher season, staffing was reduced to a few hundred servants and a few religious specialists focused on maintenance alone.[29]

Studies show that, according to their skeletal remains, most people who lived there were immigrants from diverse backgrounds. They lacked the chemical markers and osteological markers they would have if they had been living there their entire lives. Instead, research into skeletal remains has found bone damage from various species of water parasites indigenous to different areas of Peru. There were also varying osteological stressors and varying chemical densities suggesting varying long-term diets characteristic of specific regions that were spaced apart.[30] These diets are composed of varying levels of maize, potatoes, grains, legumes, and fish, but the last-known short-term diet for these people was overall composed of less fish and more corn. This suggests that several of the immigrants were from more coastal areas and moved to Machu Picchu where corn was a larger portion of food intake.[28] Most skeletal remains found at the site had lower levels of arthritis and bone fractures than those found in most sites of the Inca Empire. Incan individuals who had arthritis and bone fractures were typically those who performed heavy physical labor (such as the Mit'a) or served in the Inca military.[31]

Animals are also suspected to have been brought to Machu Picchu, as there were several bones found that were not native to the area. Most animal bones found were from llamas and alpacas. These animals naturally live at altitudes of 4,000 meters (13,000 ft) rather than the 2,400 meters (7,900 ft) elevation of Machu Picchu. Most likely, these animals were brought in from the Puna region[32] for meat consumption and for their pelts. Guinea pigs were also found at the site in special tomb caves, suggesting that they were at least used for funerary rituals,[33] as it was common throughout the Inca Empire to use them for sacrifices and meat.[34] Six dogs were also recovered from the site. Due to their placements among the human remains, it is believed that they served as companions of the dead.[33]

Agricultural activity

Andenes (terraces) used for farming at Machu Picchu

Much of the farming done at Machu Picchu was done on its hundreds of *andenes* (man-made terraces). These terraces were a work of considerable engineering, built to ensure good drainage and soil fertility while also protecting the mountain itself from erosion and landslides. However, the terraces were not perfect, as studies of the land show that there were landslides that happened during the construction of Machu Picchu. Still visible are places where the terraces were shifted by landslides and then stabilized by the Inca as they continued to build around the area.[35] Terraces constructed overlooking the Urubamba River and many other springs provided fresh water for crop production and served more than 1,000 households.[36]

However, terrace farming area makes up only about 4.9 ha (12 acres) of land, and a study of the soil around the terraces showed that what was grown there was mostly corn and potatoes, which was not enough to support the 750+ people living at Machu Picchu. This explains why studies done on the food that the Inca ate at Machu Picchu suggest it was imported from the surrounding valleys and farther afield.[30]

It is estimated that the area around the site has received more than 1,800 mm (71 in) of rain per year since AD 1450, which was more than that needed to support crop growth. Because of the ample rainfall at Machu Picchu, it was found that irrigation was not usually needed for the terraces. The terraces received so much rain that they were built by Incan engineers specifically to allow for drainage of excess water. Excavation and soil analyses done by Kenneth Wright[37][38] in the 1990s showed that the terraces were built in layers, with a bottom layer of larger stones covered by loose gravel.[35] On top of the gravel was a layer of mixed sand and gravel packed together, with rich topsoil covering it. Research showed that the topsoil was probably moved from the valley floor to the terraces because it was much better than the soil higher up the mountain.[39]

Human sacrifice and mysticism

Little information exists surrounding human sacrifices at Machu Picchu, though it is known that many sacrifices were never given a proper burial, and their skeletal remains succumbed to the elements.[40] However, there is evidence that retainers were sacrificed to accompany a deceased noble in the afterlife.[40]:107,119 Animal, liquid and dirt sacrifices to the gods were more common and were made at the Altar of the Condor. The tradition is upheld by members of the New Age Andean religion.[41]:263

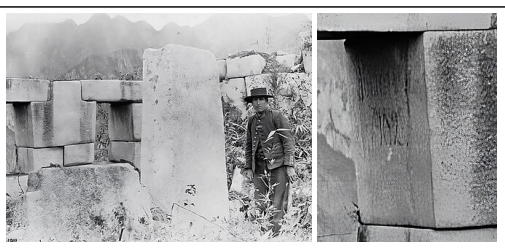
Encounters with Westerners

Spanish conquest

In the late 16th century, Spaniards who had recently gained control of the area documented that indigenous individuals mentioned returning to "Huayna Picchu", the name that is believed to be originally given to the site by locals.[15] The Spanish conquistador Baltasar de Ocampo had notes of a visit during the end of the 16th century to a mountain fortress called *Pitcos* with sumptuous and majestic buildings, erected with great skill and art, all the lintels of the doors, as well the principal as the ordinary ones, being of marble and elaborately carved.[42]

Over the centuries, the surrounding jungle overgrew the site, and few outside the immediate area knew of its existence. The site may have been re-discovered and plundered in 1867 by a German businessman, Augusto Berns.[43] Some evidence indicates that the German engineer J. M. von Hassel arrived earlier. Maps show references to Machu Picchu as early as 1874.[44] A 1904 atlas designated the site as Huayna Picchu.[15]

Search for the Neo-Inca capital

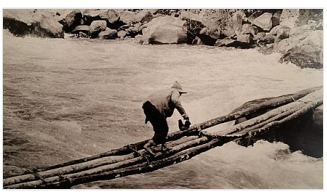


Inscription "A. Lizárraga 1902" by Lizárraga on the central window of the Temple of the Three Windows

In 1911 American historian and explorer Hiram Bingham traveled the region looking for the lost capital of the Neo-Inca state (later established to be Vilcabamba), established by Manco Inca after the Spanish conquest, and was led to Machu Picchu by a villager, Melchor Arteaga. Bingham found the name Agustín Lizárraga and the date 1902 written in charcoal on one of the walls of the Temple of the Three Windows. Initially disappointed, he documented in his pocket field journal: "Agustín Lizárraga is discoverer of Machu Picchu and lives at San Miguel Bridge just before passing."[45][46] However, while Bingham initially acknowledged Lizárraga as the discoverer in his early writings and speeches, including *Inca Land* (1922), he gradually downplayed Lizárraga's role until, in his final version of the story, *Lost City of the Incas* (1952), Bingham claimed to have found the site himself.[47]

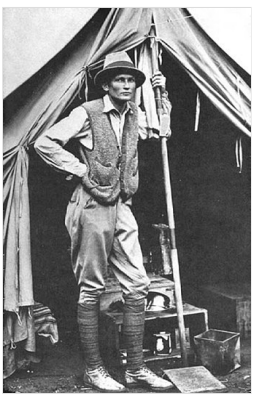
Though Bingham was not the first to visit the ruins, he was considered the scientific discoverer who brought Machu Picchu to international attention. Bingham organized another expedition in 1912 to undertake major clearing and excavation.[25]:xxx–xxxi[48]

First American expedition



Melchor Arteaga crossing the Urubamba River on 24 July 1911

Bingham was a lecturer at Yale University, although not a trained archaeologist. In 1909, returning from the Pan-American Scientific Congress in Santiago, he travelled through Peru and was invited to explore the Inca ruins at Choqquequirau in the Apurímac Valley. He organized the 1911 Yale Peruvian Expedition in part to search for the Inca capital, which was thought to be the city of Vitcos. He consulted Carlos Romero, one of the chief historians in Lima who showed him helpful references and Father Antonio de la Calancha's Chronicle of the Augustinians. In particular, Ramos thought Vitcos was "near a great white rock over a spring of fresh water." Back in Cusco again, Bingham asked planters about the places mentioned by Calancha, particularly along the Urubamba River. According to Bingham, "one old prospector said there were interesting ruins at Machu Picchu," though his statements "were given no importance by the leading citizens." Only later did Bingham learn that Charles Wiener had also heard of the ruins at Huayna Picchu and Machu Picchu, but was unable to reach them.[25][49]



Hiram Bingham III at his tent door near Machu Picchu in 1912

Armed with this information, Bingham's expedition went down the Urubamba River. En route, Bingham asked local people to show them Inca ruins, especially any place described as having a white rock over a spring.[25]:137[50]

At Mandor Pampa, Bingham asked farmer and innkeeper Melchor Arteaga if he knew of any nearby ruins. Arteaga said he knew of excellent ruins on the top of Huayna Picchu.[51] The next day, 24 July, Arteaga led Bingham and Sergeant Carrasco across the river on a log bridge and up the Machu Picchu site. At the top of the mountain, they came across a small hut occupied by a couple of Quechua people, Richard and Alvarez, who were farming some of the original Machu Picchu agricultural terraces that they had cleared four years earlier. Alvarez's 11-year-old son, Pablito, led Bingham along the ridge to the main ruins.[52]

The ruins were mostly covered with vegetation except for the cleared agricultural terraces and clearings used by the farmers as vegetable gardens. Because of the vegetation, Bingham was not able to observe the full extent of the site. He took preliminary notes, measurements, and photographs, noting the fine quality of Inca stonework of several principal buildings. Bingham was unsure about the original purpose of the ruins, but concluded there was no indication that it matched the description of Vitcos.[25]:141,186–187[50]

The expedition continued down the Urubamba and up the Vilcabamba Rivers examining all the ruins they could find. Guided by locals, Bingham rediscovered and correctly identified the site of the old Inca capital, Vitcos (then called Rosaspata), and the nearby temple of Chuquipalta. He then crossed a pass and into the Pampaconas Valley where he found more ruins heavily buried in the jungle undergrowth at Espíritu Pampa, which he named "Trombone Pampa".[53] As was the case with Machu Picchu, the site was so heavily overgrown that Bingham could only note a few of the buildings. In 1964, Gene Savoy further explored the ruins at Espiritu Pampa and revealed the full extent of the site, identifying it as Vilcabamba Viejo, where the Incas fled after the Spanish drove them from Vitcos.[54][25]:xxxv[55]

Discovery plaques located near the entrance of Machu Picchu

Bingham returned to Machu Picchu in 1912 under the sponsorship of Yale University and National Geographic and with the full support of Peruvian President Augusto Leguia. The expedition undertook a four-month clearing of the site with local labor, which was expedited with the support of the Prefect of Cusco. Excavation started in 1912 with further excavation undertaken in 1914 and 1915. Bingham focused on Machu Picchu because of its fine Inca stonework and well-preserved nature, which had lain undisturbed since the site was abandoned. None of Bingham's several hypotheses explaining the site have since held up. During his studies, he carried various artifacts back to Yale. One prominent artifact was a set of 15th-century, ceremonial Incan knives made from bismuth bronze; they are the earliest known artifact containing this alloy.[56][57]

Although local institutions initially welcomed the exploration, they soon accused Bingham of legal and cultural malpractice.[58] Rumors arose that the team was stealing artifacts and smuggling them out of Peru through Bolivia. In fact, Bingham removed many artifacts, but openly and legally; they were deposited in the Yale University Museum. Bingham was abiding by the 1852 Civil Code of Peru; the code stated that "archaeological finds generally belonged to the discoverer, except when they had been discovered on private land" (Batievsky 100).[59] Local press perpetuated the accusations, claiming that the excavation harmed the site and deprived local archaeologists of knowledge about their own history.[58] Landowners began to demand rent from the excavators.[58] By the time Bingham and his team left Machu Picchu, locals had formed coalitions to defend their ownership of Machu Picchu and its cultural remains, while Bingham claimed the artifacts ought to be studied by experts in American institutions.[58]

Current state

Preservation

In 1981, Peru declared an area of 325.92 square kilometers (125.84 sq mi) surrounding Machu Picchu a "historic sanctuary".[60] In addition to the ruins, the sanctuary includes a large portion of the adjoining region, rich with the flora and fauna of the Peruvian Yungas and Central Andean wet puna ecoregions.[61]

Beyond its historical significance, Machu Picchu houses a diverse range of species. Among them are the Andean fox, puma, vizcacha, spectacled bear, and white-tailed deer. The sanctuary is also habitat for more than 420 bird species, notably the cock-of-the-rock and the Andean condor. The surrounding environment features a variety of tree species such as alder, white cedar, husk, and laurel.[62]

In 1983, UNESCO designated Machu Picchu a World Heritage Site, describing it as "a masterpiece of art, urbanism, architecture and engineering" and "a unique testimony" of the Inca Civilization.[8]

The modern town of Machu Picchu

Main Square of Aguas Calientes

Along the Urubamba river, below the ruins, surrounding the train line "street", is the town of Machu Picchu, also known as Aguas Calientes (hot springs), with a post office, a train station, hotels, and other services for the many tourists. The station, called Puente Ruinas (the bridge to the ruins) is the end of the line for the *tren de turismo*, the tourist train, which arrives every morning from Cusco and returns every afternoon. There is also a luxury hotel on the mountain, near the ruins.[63]

Machu Picchu is officially twinned with Haworth, West Yorkshire in the United Kingdom,[64] as well as Fukushima[65] and Ōtama, Japan,[66] Petra, Jordan,[67] Medley, United States,[68] and Tinum, Mexico.[69]

Tourist activity

Machu Picchu is both a cultural and natural UNESCO World Heritage Site. Since Hiram Bingham's expedition, growing numbers of tourists have visited the site each year, with numbers exceeding 1.4 million in 2017.[70] As Peru's most visited tourist attraction, and a major revenue generator, it is continually exposed to economic and commercial forces. In the late 1990s, the Peruvian government granted concessions to allow the construction of a cable car and a luxury hotel, including a tourist complex with boutiques and restaurants and a bridge to the site.[71] Many people protested the plans, including Peruvians and foreign scientists, saying that more visitors would pose a physical burden on the ruins.[72] In 2018, plans were restarted to again construct a cable car to encourage Peruvians to visit Machu Picchu and boost domestic tourism.[73] A no-fly zone exists above the area.[74] UNESCO decided not to include Machu Picchu on its List of World Heritage in Danger in 2017.[75][76]

Tourists at the main entrance to the Machu Picchu citadel

Efforts to manage the impact of tourism have included various measures over the years. In the 1980s, a rock from Machu Picchu's central plaza was moved to create a helicopter landing zone, a practice which was later stopped. In 2006, the company Helicusco sought approval for tourist flights over the site, but the license was soon rescinded.[74] In January 2010, severe flooding caused by El Niño trapped over 4,000 people and disrupted access to Machu Picchu, leading to its temporary closure.[77] The site reopened on April 1, 2010.[78] To further address tourism's impact, stricter entrance regulations were introduced in July 2011, limiting the number of daily visitors to 2,500 per day to the citadel and 400 to Huayna Picchu.[79] In 2018, a third entrance phase was added to better manage tourism and reduce site degradation.[80]

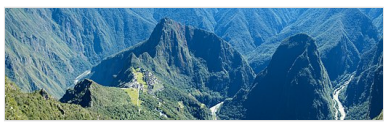
In May 2012, UNESCO urged additional protection measures for the site's buffer zone, especially due to the rapid growth in the nearby town of Aguas Calientes.[81] Tourist deaths at Machu Picchu from altitude sickness, floods, and accidents have led to criticism of UNESCO for allowing visits despite the site's high safety risks.[82][83][84][85]

The trend of nude tourism in 2014 also led to increased surveillance by Peru's Ministry of Culture to end the practice.[86]

From 1994 to 2019, Fernando Astete served as the Chief of the National Archaeological Park of Machu Picchu.[87] His over 30 years of work in preservation and research significantly advanced the scientific understanding of the site and promoted sustainable tourism practices. His efforts led to a greater appreciation of the Inca landscape among the public and the scientific community.[87]

During the 2022–2023 Peruvian protests, routes to Machu Picchu were blocked, trapping thousands of tourists and leading to a government airlift of the stranded visitors.[88][89] Due to these disruptions, the Ministry of Culture closed the site indefinitely on January 22, 2023, and it was reopened on February 15, 2023.[88][90][91]

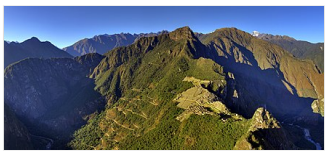
Geography



Panoramic view of Machu Picchu from Machu Picchu mountain surrounded by the Urubamba River

Machu Picchu lies in the Southern Hemisphere, 13.111 degrees south of the equator.[92] It is 80 kilometers (50 miles) northwest of Cusco, on the crest of the mountain Machu Picchu, located about 2,430 meters (7,970 feet) above mean sea level, over 1,000 meters (3,300 ft) lower than Cusco, which has an elevation of 3,400 meters (11,200 ft).[92] As such, it had a milder climate than the Inca capital. It is one of the most important archaeological sites in South America, one of the most visited tourist attractions in Latin America and the most visited in Peru.[93]

Machu Picchu is situated above a bow of the Urubamba River, which surrounds the site on three sides, where cliffs drop vertically for 450 meters (1,480 ft) to the river at their base. The area is subject to morning mists rising from the river.[52] When inhabited by the Incas, the location of the city was a military secret, and its deep precipices and steep mountains provided natural defenses. The Inca Bridge, an Inca grass rope bridge, across the Urubamba River in the Pongo de Mainique, provided a secret entrance for the Inca army. Another Inca bridge was built to the west of Machu Picchu, the tree-trunk bridge, at a location where a gap occurs in the cliff that measures 6 meters (20 ft).



Machu Picchu as seen from Wayna Picchu

The city sits in a saddle between the two mountains Machu Picchu and Huayna Picchu,[52] with a view down two valleys and a nearly impassable mountain at its back. It has a water supply from springs that cannot be easily blocked. The Incas used sloped hillsides to their benefit, using terraces to provide more farmland, grow crops, and steepen the slopes that invaders would have to ascend. The terraces also reduced soil erosion and protected against landslides.[94] Two high-altitude routes from Machu Picchu cross the mountains back to Cusco, one through the Sun Gate, and the other across the Inca bridge. Both could have been blocked easily, should invaders have approached along them.

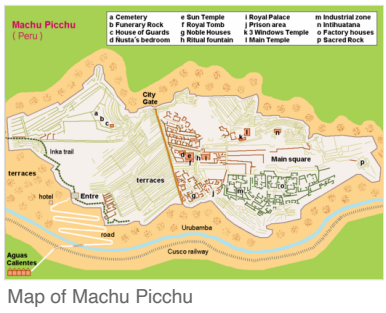
Machu Picchu and other sites in the area are built over earthquake faults. According to research conducted in 2019, this may not be a coincidence: "[o]ne simple answer, researchers now suggest, is that that's [earthquake faults] where building materials for the site—large amounts of already fractured rock—were readily available."[95]

Climate

Between the valley floor and the altitudinal zone of the Inca citadel, ranging from 2,200 metres (7,200 ft) to 2,500 metres (8,200 ft) meters above sea level, the climate is warm and humid, with an average annual precipitation of 2,010 millimetres (79 in) and an annual mean temperature of approximately 18 °C (64 °F). The site is characterized by steep slopes, dense vegetation, and significant rainfall, contributing to high humidity levels of 80–90%. The area is also frequently enveloped in cloud cover, typical of a cloud forest environment.[96]

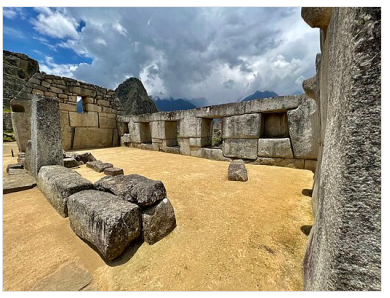
Climate data for Machu Picchu (elevation 2,399 m (7,871 ft), 1991–2020 normals)													
Month	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**	**Year**
Mean daily maximum °C (°F)	20.6 (69.1)	20.6 (69.1)	21.0 (69.8)	21.5 (70.7)	21.9 (71.4)	22.0 (71.6)	22.1 (71.8)	23.1 (73.6)	23.4 (74.1)	22.8 (73.0)	22.8 (73.0)	21.2 (70.2)	21.9 (71.5)
Mean daily minimum °C (°F)	11.8 (53.2)	11.9 (53.4)	11.9 (53.4)	11.7 (53.1)	10.8 (51.4)	10.0 (50.0)	9.4 (48.9)	10.0 (50.0)	11.1 (52.0)	11.5 (52.7)	12.0 (53.6)	12.0 (53.6)	11.2 (52.1)
Average precipitation mm (inches)	340.7 (13.41)	324.5 (12.78)	340.6 (13.41)	184.1 (7.25)	73.4 (2.89)	50.8 (2.00)	54.2 (2.13)	60.6 (2.39)	76.2 (3.00)	163.0 (6.42)	172.5 (6.79)	275.4 (10.84)	2,116 (83.31)
Source: National Meteorology and Hydrology Service of Peru[97]													

Site



Map of Machu Picchu

Layout



Temple of the Three Windows

The site is roughly divided into an urban sector and an agricultural sector, with the former divided between an upper town and a lower town. The temples are in the upper town, the warehouses in the lower.[98]

The architecture is adapted to the mountains. Approximately 200 buildings are arranged on wide parallel terraces around an east–west central square. The various compounds, called *kanchas*, are long and narrow in order to exploit the terrain. Sophisticated channeling systems provided irrigation for the fields. Stone stairways set in the walls allowed access to the different levels across the site. The eastern section of the city was probably residential. The western, separated by the square, was for religious and ceremonial purposes. This section contains the *Torreón*, the massive tower which may have been used as an observatory.[99]

Located in the urban sector are the primary archaeological treasures and tourist attractions: the *Intihuatana*, the *Temple of the Sun* and the *Temple of the Three Windows*.[100]

The Popular District, or Residential District, is the place where the lower-class people lived. It includes storage buildings and simple houses.[101]

The royalty area, a sector for the nobility, is a group of houses located in rows over a slope; the residence of the *amautas* (wise people) was characterized by its reddish walls, and the zone of the *ñustas* (princesses) had trapezoid-shaped rooms. The Monumental Mausoleum is a carved statue with a vaulted interior and carved drawings. It was used for rites or sacrifices.

The Guardhouse is a three-sided building, with one of its long sides opening onto the Terrace of the Ceremonial Rock. The three-sided style of Inca architecture is known as the *wayrona* style.[102]

In 2005 and 2009, the University of Arkansas made detailed laser scans of the entire site and of the ruins at the top of the adjacent Huayna Picchu mountain. The scan data is available online for research purposes.[103]

Sites of interest

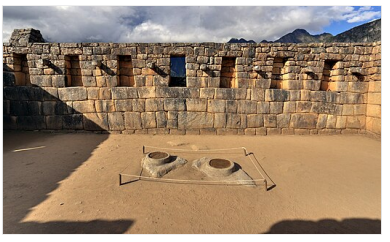
Temple of the Sun or Torreón

Temple of the Sun or Torreon

This semicircular temple is built on the same rock overlying Bingham's "Royal Mausoleum", and is similar to the Temple of the Sun found in Cusco and the one found in Pisac, in having what Bingham described as a "parabolic enclosure wall". The stonework is of ashlar quality. Within the temple is a 1.2 m by 2.7 m rock platform, smooth on top except for a small platform on its southwest quadrant. A "Serpent's Door" faces 340°, or just west of north, opening onto a series of 16 pools, and affording a view of Huayna Picchu. The temple also has two trapezoidal windows, one facing 65°, called the "Solstice Window", and the other facing 132°, called the "Qullqa Window". The northwest edge of the rock platform points out the Solstice Window to within 2' of the 15th century June solstice rising Sun. For comparison, the angular diameter of the Sun is 32'. The Inca constellation Qullca, storehouse, can be viewed out the Qullqa Window at sunset during the 15th-century June Solstice, hence the window's name. At the same time, the Pleaides are at the opposite end of the sky. Also seen through this window on this night are the constellations Llamacnawin, Llama, Unallamacha, Machacuay, and the star Pachapacariq Chaska (Canopus).[49][104]

Intihuatana is believed to have been designed as an astronomic clock or calendar by the Incas

Intihuatana stone



The sculpture projecting from the rock bottom of the Sun temple is interpreted as "Water mirrors for observing the sky".[105]

The *Intihuatana* stone is a ritual stone which was used by the Incas. The stone is designed to point directly at the Sun during the winter solstice.[106] The name of the stone (perhaps coined by Bingham) derives from Quechua language: *inti* means "sun", and *wata-*, "to tie, hitch (up)". The suffix *-na* derives nouns for tools or places. Hence *Intihuatana* is literally an instrument or place to "tie up the sun", often expressed in English as "The Hitching Post of the Sun". The Inca believed the stone held the Sun in its place along its annual path in the sky.[107] The

stone is situated at 13°9'48" S. At midday on 11 November and 30 January, the Sun is situated almost exactly above the pillar, casting no shadow. On 21 June, the stone casts the longest shadow on its southern side, and on 21 December a much shorter shadow on its northern side.[108]

Inti Mach'ay and the Royal Feast of the Sun

Inti Mach'ay is a special cave used to observe the Royal Feast of the Sun. This festival was celebrated during the Incan month of *Qhapaq Raymi*. It began earlier in the month and concluded on the December solstice. On this day, noble boys were initiated into manhood by an ear-piercing ritual as they stood inside the cave and watched the sunrise.[109]

Architecturally, Inti Mach'ay is often considered to be the most significant structure at Machu Picchu. Its entrances, walls, steps, and windows are some of the finest masonry in the Incan Empire. The cave also includes a tunnel-like window unique among Incan structures, which was constructed to allow sunlight into the cave only during several days around the December solstice. For this reason, the cave was inaccessible for much of the year.[110] Inti Mach'ay is located on the eastern side of Machu Picchu, just north of the "Condor Stone." Many of the caves surrounding this area were prehistorically used as tombs, yet there is no evidence that Mach'ay was a burial ground.[111]

Dispute over cultural artifacts

Artifact collected on Bingham's 1912 expedition, on display at the Museo Machu Picchu

In 1912, 1914 and 1915, Bingham removed thousands of artifacts from Machu Picchu—ceramic vessels, silver statues, jewelry, and human bones—and took them to Yale University for further study, supposedly for 18 months. Yale instead kept the artifacts until 2012, arguing that Peru lacked the infrastructure and systems to care for them. Eliane Karp, an anthropologist and wife of former Peruvian President Alejandro Toledo, accused Yale of profiting from Peru's cultural heritage. Many of the articles were exhibited at Yale's Peabody Museum.[112]

Aryballos and bowls returned by Yale University in 2011

In 2006, Yale returned some pieces but retained more than 250 "museum-quality" pieces, claiming this was supported by federal case law of Peruvian antiquities.[113] In 2007, the Peruvian government and Yale had agreed on a joint traveling exhibition and construction of a new museum and research center in Cusco advised by Yale. Yale acknowledged Peru's title to all the objects, but would share rights with Peru in the research collection, part of which would remain at Yale for continuing study.[114] In November 2010, Yale agreed to return the disputed artifacts.[115] The third and final batch of artifacts was delivered in November 2012.[116] The artifacts are permanently exhibited at the Museo Machu Picchu, La Casa Concha ("The Shell House"), close to Cusco's colonial center. Owned by the National University of San Antonio Abad del Cusco, La Casa Concha also features a study area for local and foreign students.[117]

Construction

View of the residential section of Machu Picchu

Funerary Stone in upper cemetery

The central buildings in the Machu Picchu citadel use the classical Inca architectural style of polished dry-stone walls of regular shape. The Incas were masters of this technique, called ashlar, in which blocks of stone are cut to fit together tightly without mortar.

The site itself may have been intentionally built on fault lines to afford better drainage and a ready supply of fractured stone, with one article in ScienceDaily stating "Machu Picchu clearly shows us that the Incan civilization was an empire of fractured rocks".[118]

The section of the mountain where Machu Picchu was built provided various challenges that the Incas solved with local materials. One issue was the seismic activity due to two fault lines which made mortar and similar building methods nearly useless. Instead, the Inca mined stones from some quarries at the site,[119] including one recently discovered using remote sensing techniques, which was probably located in the catchment area between the Hurin and Hanan before it was covered over to create the current Plaza Principal.[120] Once mined, the Inca lined the granite stones up and shaped them to fit together perfectly, stabilizing the structures. Inca walls have many stabilizing features: doors and windows are trapezoidal, narrowing from bottom to top; corners usually are rounded; inside corners often incline slightly into the rooms, and outside corners were often tied together by L-shaped blocks; walls are offset slightly from row to row rather than rising straight from bottom to top.[121]

Heavy rainfall required terraces to drain rain water and prevent mudslides, landslides, erosion, and flooding. Terraces were layered with stone chips, sand, dirt, and topsoil, to absorb water and prevent it from running down the mountain. Similar layering protected the large city center from flooding.[122] Multiple canals and reserves throughout the city provided water that could be supplied to the terraces for irrigation and to prevent erosion and flooding.[123]

The approach to moving and placing the enormous stones remains uncertain, probably involving hundreds of men to push the stones up inclines. A few stones have knobs that could have been used to lever them into position; the knobs were generally sanded away, with a few overlooked.[124] The Incas never used wheels in a practical way, although their use in toys show that they knew the principle. The use of wheels in engineering may have been limited due to the lack of strong draft animals, combined with steep terrain and dense vegetation.[125]

Transportation

The Inca road system included a route to the Machu Picchu region. The people of Machu Picchu were connected to long-distance trade, as shown by non-local artifacts found at the site. For example, Bingham found unmodified obsidian nodules at the entrance gateway. In the 1970s, Burger and Asaro determined that these obsidian samples were from the *Titicaca* or Chivay obsidian source, and that the samples from Machu Picchu showed long-distance transport of this obsidian type in pre-Hispanic Peru.[126]

View of Machu Picchu arriving from the Inca Trail

Today, visitors can reach Machu Picchu via three main routes. The most popular option is the train journey from Cusco or Ollantaytambo. Operated by PeruRail and Inca Rail, this scenic route takes visitors to the town of Aguas Calientes, from where they can take a 25-minute bus ride to the Machu Picchu entrance or walk 8 kilometres (5.0 mi), which takes about 1.5 hours.[127]

Another widely chosen route is the Inca Trail, a historic path built by the Incas considered "the most famous hike in South America" by Lonely Planet,[128] attracting thousands of tourists each year.[129] The trail, which starts at Piscacucho (also known as Km 82 on the railway to Aguas Calientes) or Km 104, depending on the length of the trek, spans approximately 42 kilometres (26 mi) and reaches an altitude of 4,200 metres (13,800 ft).[130][128] It passes through the Andes and various archaeological sites before reaching the Sun Gate (Intipunku) at Machu Picchu. Due to limited permits, advance booking through a tourism agency is required.[130]

Alternatively, travelers can take a less conventional route that involves traveling by road to a point near the Hydroelectric station. From there, they can either walk for about 2 hours or take a short train ride to Aguas Calientes. [131] This option is generally more affordable and does not require advance booking, but it is less recommended during the rainy season due to potential landslide risks.[132]

In media

Motion pictures

The Paramount Pictures film *Secret of the Incas* (1954), with Charlton Heston and Yma Sumac, was filmed on location at Cusco and Machu Picchu, the first time that a major Hollywood studio filmed on site. Five hundred indigenous people were hired as extras in the film.[133]

The opening sequence of the film *Aguirre, the Wrath of God* (1972) was shot in the Machu Picchu area and on the stone stairway of Huayna Picchu.[134]

Machu Picchu was featured prominently in the film *The Motorcycle Diaries* (2004), a biopic based on the 1952 youthful travel memoir of Marxist revolutionary Che Guevara.[135]

The *NOVA* television documentary "Ghosts of Machu Picchu" presents an elaborate documentary on the mysteries of Machu Picchu.[136]

Multimedia artist Kimsooja used footage shot near Machu Picchu in the first episode of her film series *Thread Routes*, shot in 2010.[137]

On their 7th release, "Transformers: Rise of the Beasts" (2023) brought the Autobots to Machu Picchu and more of Cusco's famous sites.[138]

Music

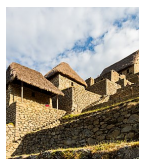
The song "Kilimanjaro", from the Indian Tamil language film *Enthiran* (2010), was filmed in Machu Picchu.[139] The sanction for filming was granted only after direct intervention from the Indian government.[140][141]

Gallery

General view.

Sunrise from Inti Punku, in the final stretch of the Inca Trail.

Detail of a trapezoidal window.



Colcas on the agricultural sector.

The Temple of the Condor.

Terraces.

Detail of the wall's masonry.

Llamas at Machu Picchu.

Residential area.

Cliff.

See also

Civilizations portal

Cities portal

Peru portal

- Iperú, tourist information and assistance
- Lares trek, an alternative route to that of the Inca Trail
- List of archaeological sites in Peru
- List of archaeoastronomical sites by country
- Paleohydrology
- Putucusi, neighboring mountain
- Religion in the Inca Empire
- Salcantay Trek – alternative trek to Machu Picchu
- The Chilean Inca Trail
- Tourism in Peru

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Notes

a. For most English or Spanish speakers, the first 'c' in *Picchu* is silent. In English, the name is pronounced /ˌmɑːtʃuː ˈpiːtʃuː/ *MAH-choo PEE-choo*[3][4] or /ˌmætʃuː ˈpiːktʃuː/ *MATCH-oo PEAK-choo*,[4][5] in Spanish as [ˈmatʃu ˈpitʃu] or [ˈmatʃu ˈpiɣtʃu],[6] and in Quechua (*Machu Pikchu*)[7] as [ˈmatʃʊ ˈpɪktʃʊ].

b. *Pikchu* may also refer to a "portion of coca that is chewed".

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External links

- Official website of the Ministry of Culture of Peru (https://www.machupicchu.gob.pe/?lang=en)
- UNESCO – Machu Picchu (World Heritage) (https://whc.unesco.org/pg.cfm?cid=31&id_site=274)
- Machu Picchu Multimedia Resources (https://www.topworldimages.com/Machu_Picchu.htm)
- Stories on Machu Picchu by Fernando Astete, former Chief of National Archaeological Park of Machupicchu (https://www.machupicc.hu/)
- Plants and animals in Machu Picchu (https://www.responsiblepeople.org/en/blog/machu-picchu-animals) Archived (https://web.archive.org/web/20231104173159/https://www.responsiblepeople.org/en/blog/machu-picchu-animals) 4 November 2023 at the Wayback Machine

Images

- First photographs of Hiram Bingham in Machu Picchu (https://historycollection.com/21-striking-photographs-machu-picchu-discovery/)
- Archive of Fernando Astete, former Chief of the National Archaeological Park of Machupicchu (https://iconoteca.arc.usi.ch/en/collection/1824)

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